

8086 Microprocessor in Maximum Mode

Q. Explain the operation of the 8086 microprocessor in Maximum Mode. Discuss the role of the Bus Controller (8288) and how a multiprocessor environment is managed.

[DEF] Definition to Remember

The 8086 operates in Maximum Mode when the MN/MX (Pin 33)' pin is tied to Ground (Logic Low). Designed for multi-processor systems (e.g., using an 8087 coprocessor), the 8086 stops generating control signals and instead sends status codes to an external 8288 Bus Controller.

1. Maximum Mode Configuration

In this mode, pins that previously generated control signals (like ALE, WR') take on new functions related to bus arbitration and multiprocessor coordination.

2. Reassigned Maximum Mode Signals

* S2', S1', S0' (Status Lines): The 8086 outputs its current cycle status on these 3 pins. The 8288 reads them to generate command signals.

3. Role of the 8288 Bus Controller

The 8288 decodes the S2'-S0' signals to generate system controls:

4. Managing the Multiprocessor Environment

* Bus Arbitration: To prevent data corruption, only one processor can use the bus at a time. A coprocessor asserts RQ'. The 8086 finishes its cycle, floats its buses (High-Z), and asserts GT' to hand over control.

[MUST-WRITE] Must-Write Points (for 10 marks)

1. Maximum Mode is activated by tying the MN/MX (Pin 33)' pin to Ground (Low).
2. Designed for multi-processor environments (e.g., CPU + 8087 Math Coprocessor).
3. The 8086 stops generating control signals; relies on the 8288 Bus Controller.
4. 8086 outputs status on S2, S1, S0, which the 8288 decodes to generate Memory/IO Read/Write signals.
5. HOLD/HLDA are replaced by bidirectional RQ'/GT' (Request/Grant) for bus arbitration.
6. LOCK' pin: Asserts exclusive control over the bus to protect shared memory.
7. QS0/QS1: Broadcast instruction queue status to coprocessors.

[RECALL] Quick Recall

Maximum Mode -> MN/MX (Pin 33)' = GND -> Multi-processor -> 8288 Bus Controller does the work -> CPU sends S0, S1, S2 -> Uses RQ/GT for bus request -> LOCK to protect shared memory